

Allocation results from manipulative experiments

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LEMONTREE Science meeting

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General approach to measuring allocation in experiments

General approach to measuring allocation in experiments

- Within leaves
 - Nitrogen allocation
 - Phosphorus allocation

General approach to measuring allocation in experiments

- Within leaves
 - Nitrogen allocation
 - Structure (LMA)
 - RuBisCO (V_{cmax})
 - Light harvesting (chlorophyll)
 - Bioenergetics (electron transport; J_{max})
 - Phosphorus allocation

General approach to measuring allocation in experiments

- Within leaves
 - Nitrogen allocation
 - Phosphorus allocation
 - Inorganic phosphate
 - Nucleic acids
 - Phospholipids
 - Metabolites
 - Residual

General approach to measuring allocation in experiments

- Within leaves
 - Nitrogen allocation
 - Phosphorus allocation
- Whole plant

General approach to measuring allocation in experiments

- Within leaves
 - Nitrogen allocation
 - Phosphorus allocation
- Whole plant
 - Root:Shoot ratio
 - Total leaf area

Allocation in manipulative contexts

- Light availability
- CO₂ availability
- Vapor pressure deficit
- Water availability
- Nitrogen availability
- Phosphorus availability
- N-fixing bacteria presence
- Mycorrhizae presence

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Many interactive combinations!

Allocation in manipulative contexts

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Light availability impacts on allocation

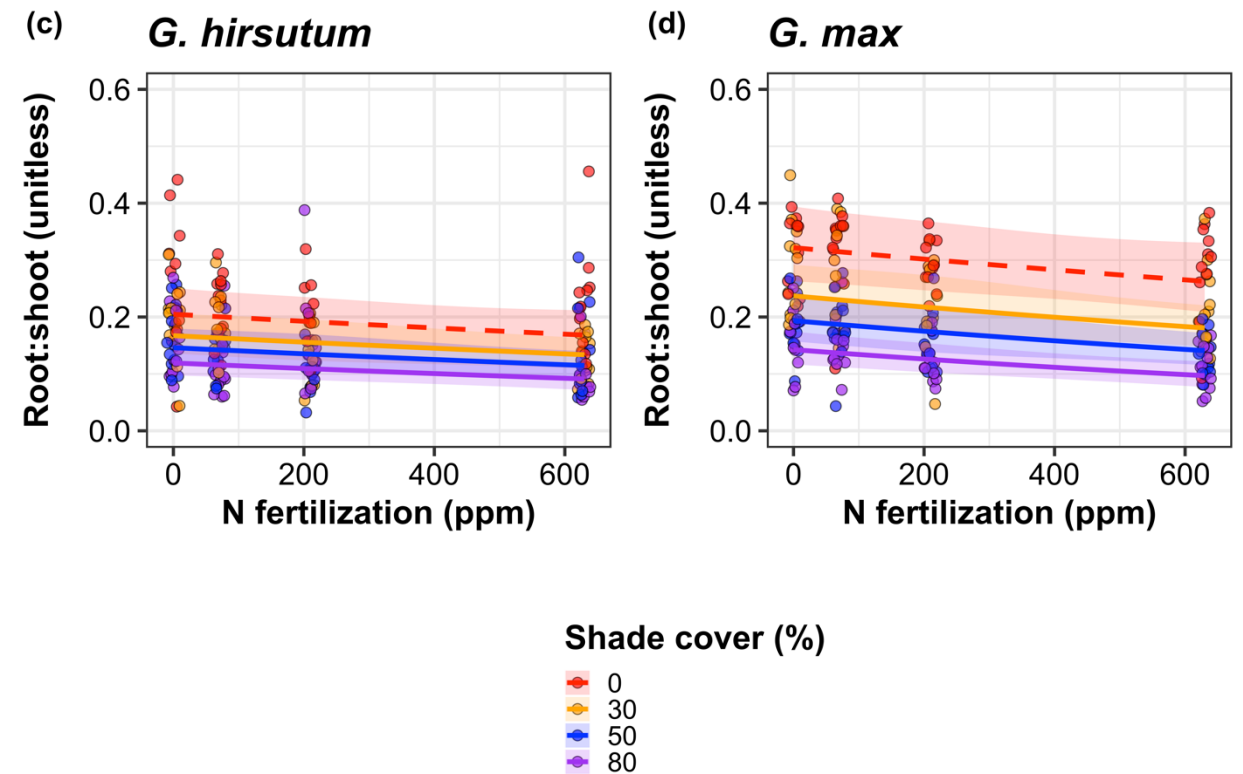
- Within leaves
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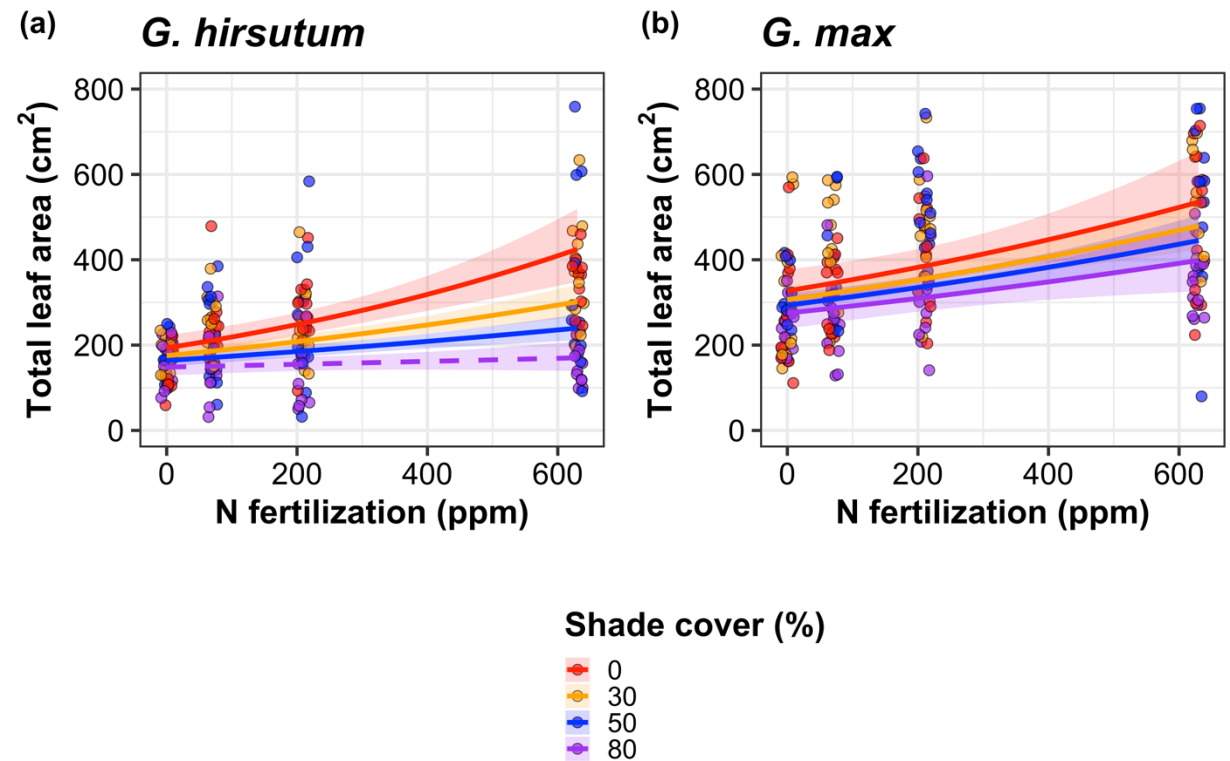
Increasing light increases R:S



Light availability impacts on allocation

- Within leaves
 - Nitrogen allocation
 - Phosphorus allocation
- Whole plant
 - Root:Shoot ratio
 - **Total leaf area**

Increasing light increases TLA



Allocation in manipulative contexts

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CO₂ availability impacts on allocation

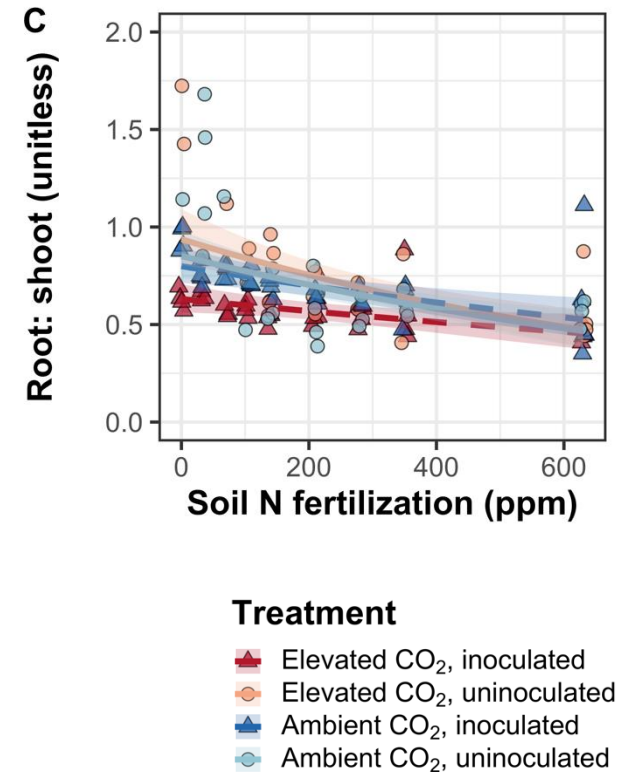
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CO₂ availability impacts on allocation

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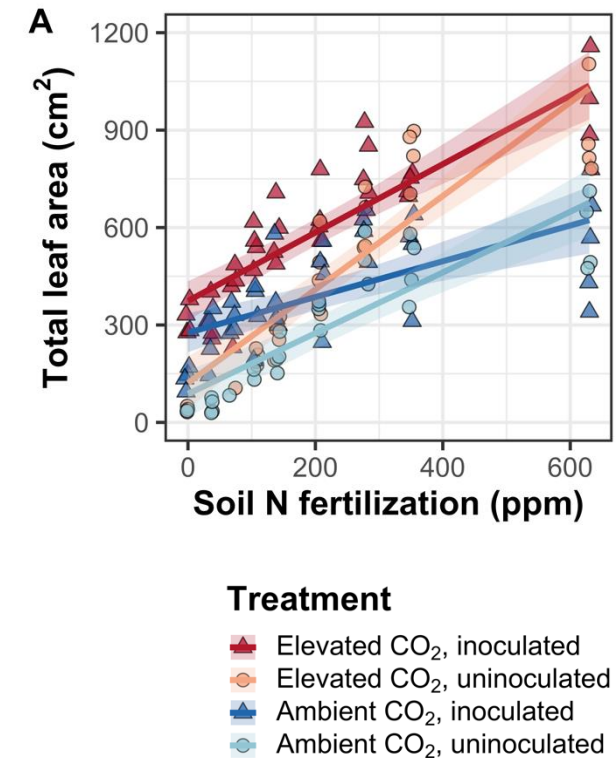
Slight R:S decrease in plants inoculated with N-fixing bacteria



CO₂ availability impacts on allocation

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TLA increased with CO₂, especially with adequate N



Allocation in manipulative contexts

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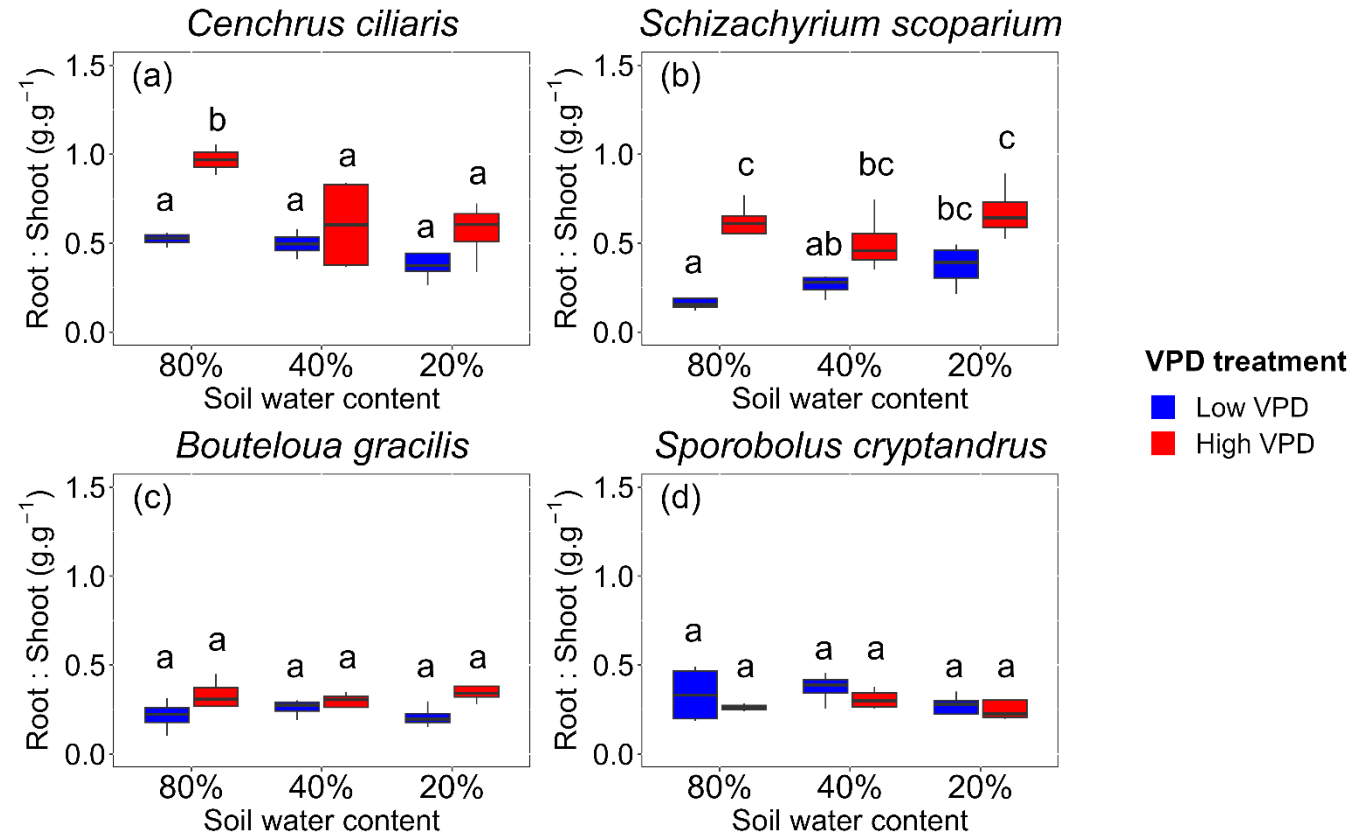
VPD impacts on allocation

- Within leaves
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VPD impacts on allocation

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VPD increases R:S in acquisitive, but not conservative species

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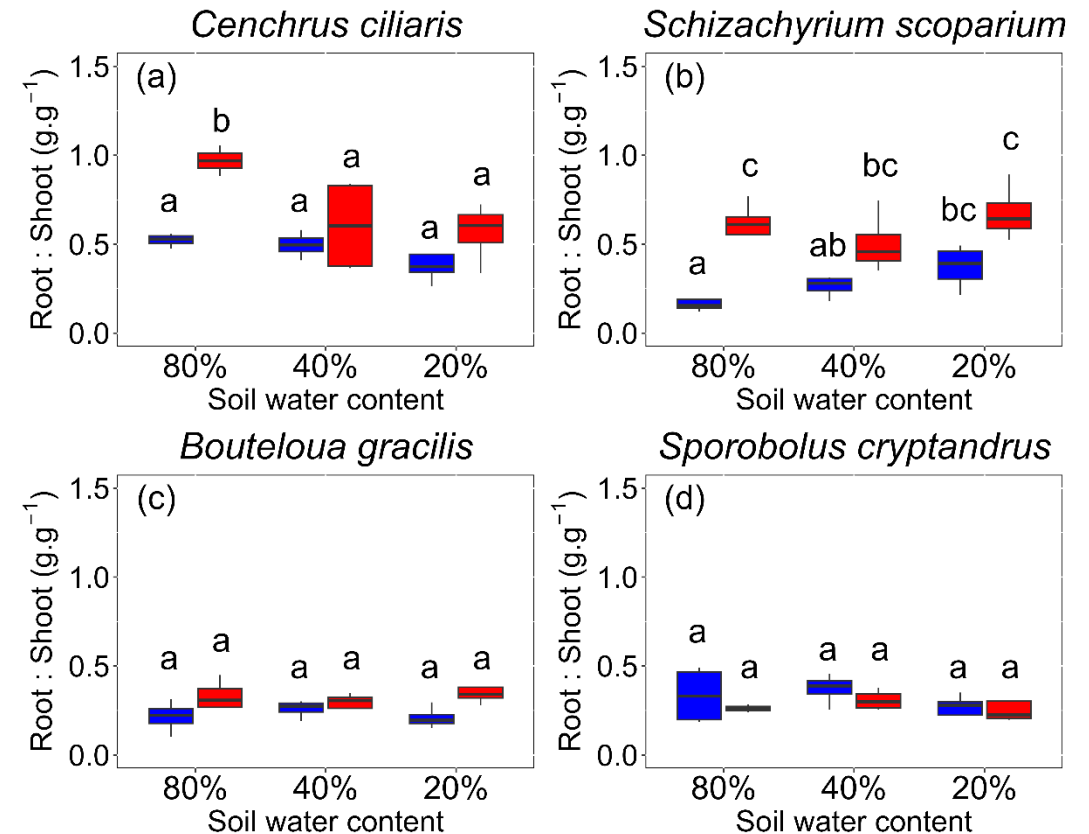
Soil water impacts on allocation

- Within leaves
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 - Phosphorus allocation
- Whole plant
 - Root:Shoot ratio
 - Total leaf area



Soil water impacts on allocation

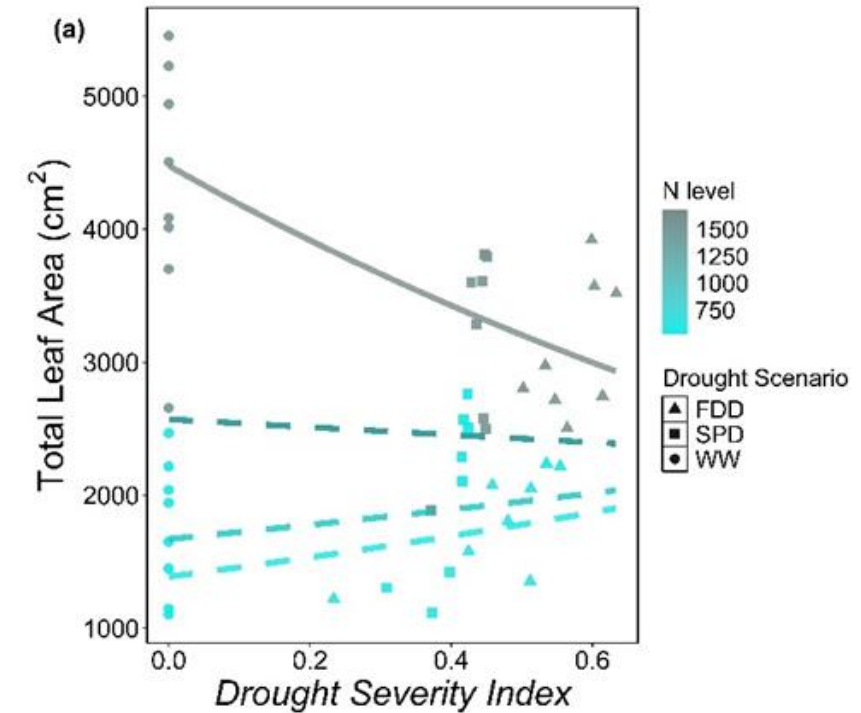
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Water availability not impacting R:S in our experiments

Soil water impacts on allocation

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Water availability increases TLA when N is available

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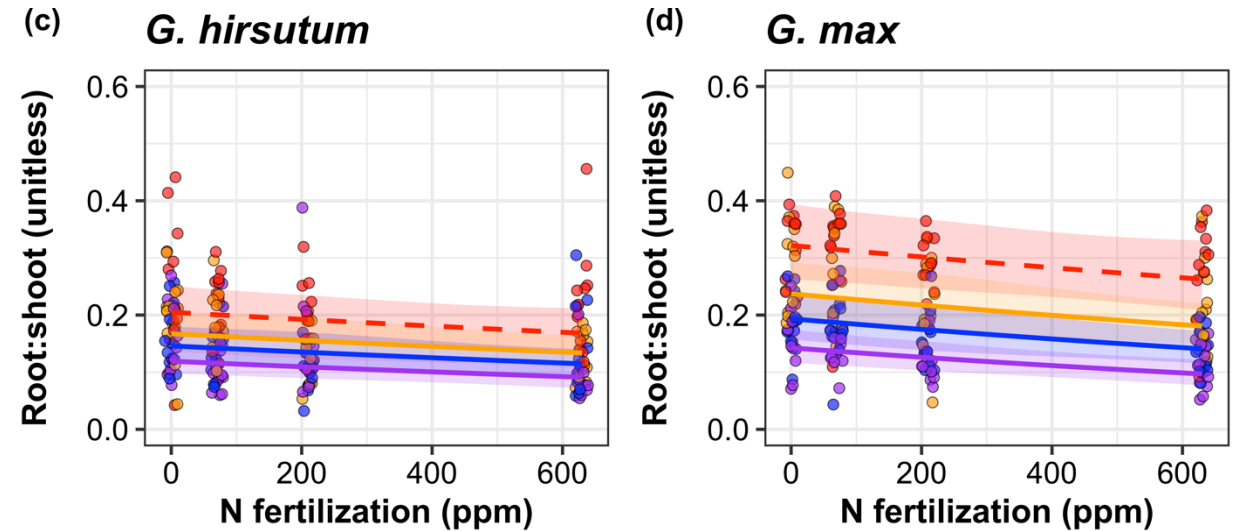
Soil nitrogen impacts on allocation

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Soil nitrogen impacts on allocation

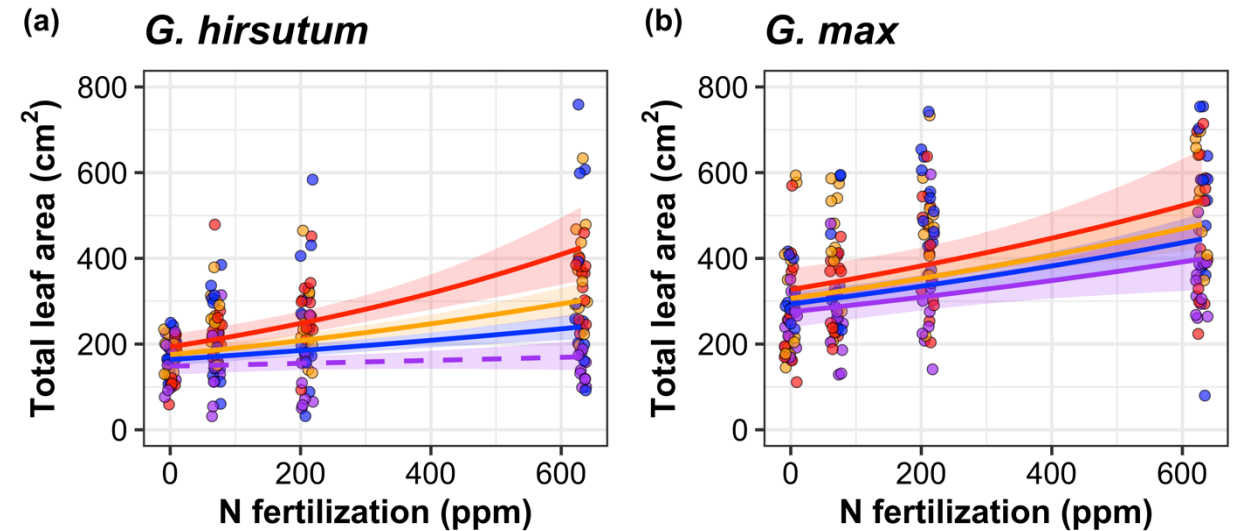
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Soil N decreases R:S

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Soil N increases TLA

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Soil phosphorus impacts on allocation

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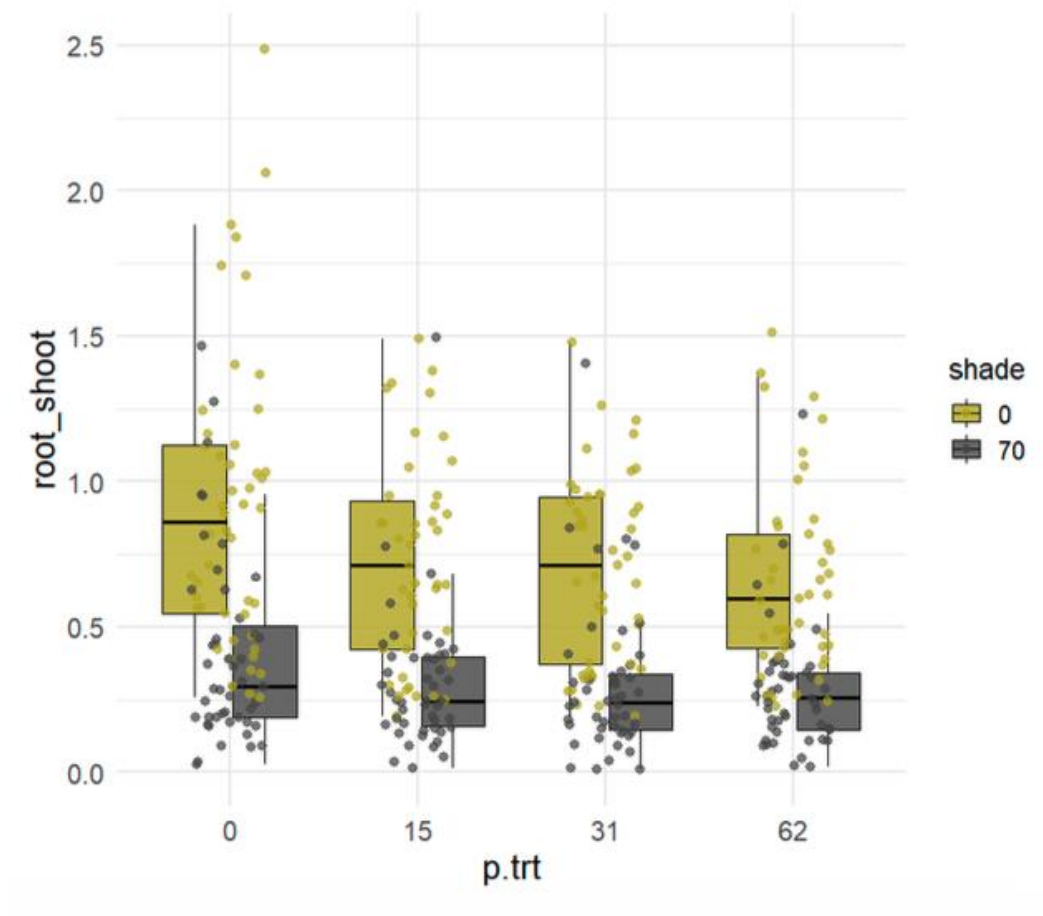
P allocation data being collected now!



Soil phosphorus impacts on allocation

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Soil P availability decreases R:S



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Happy to chat more about these later

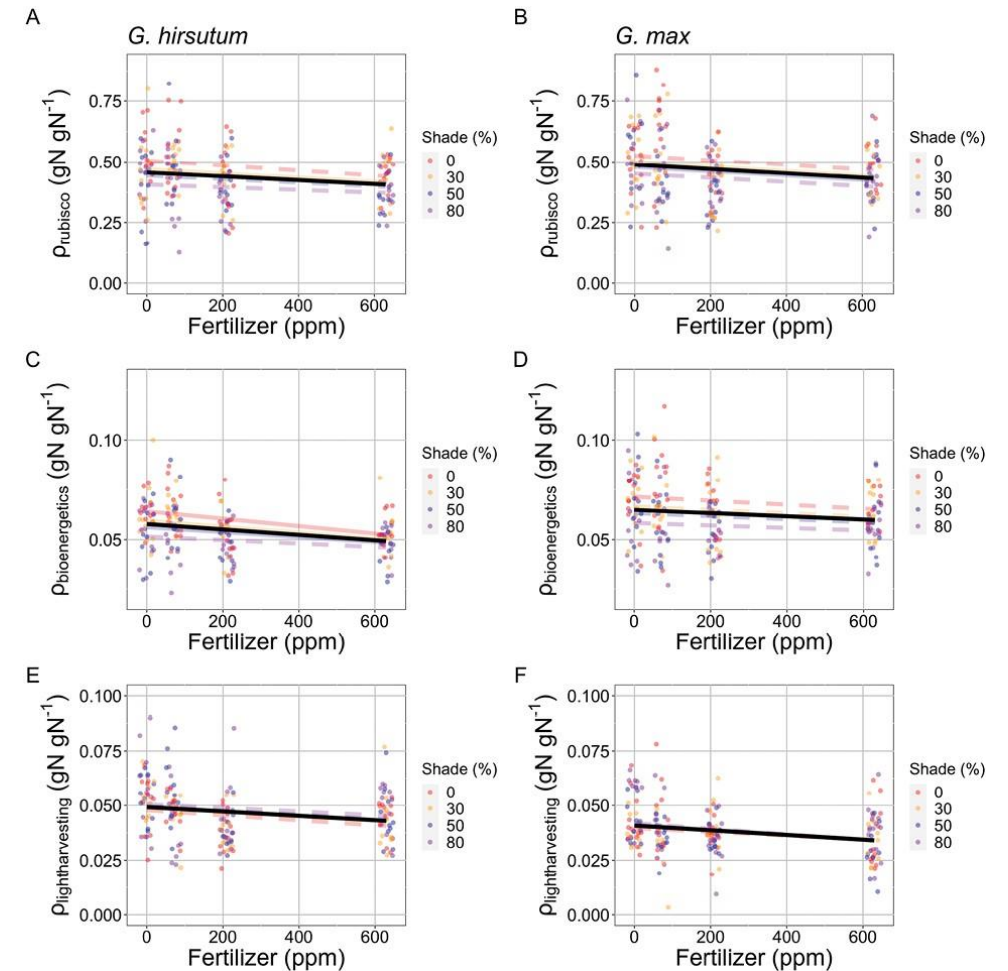
Conclusions

- R:S increases with increasing soil resource demand and decreasing soil resource availability
- TLA increases with resource demand and availability

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Light increases leaf N allocation to photosynthesis



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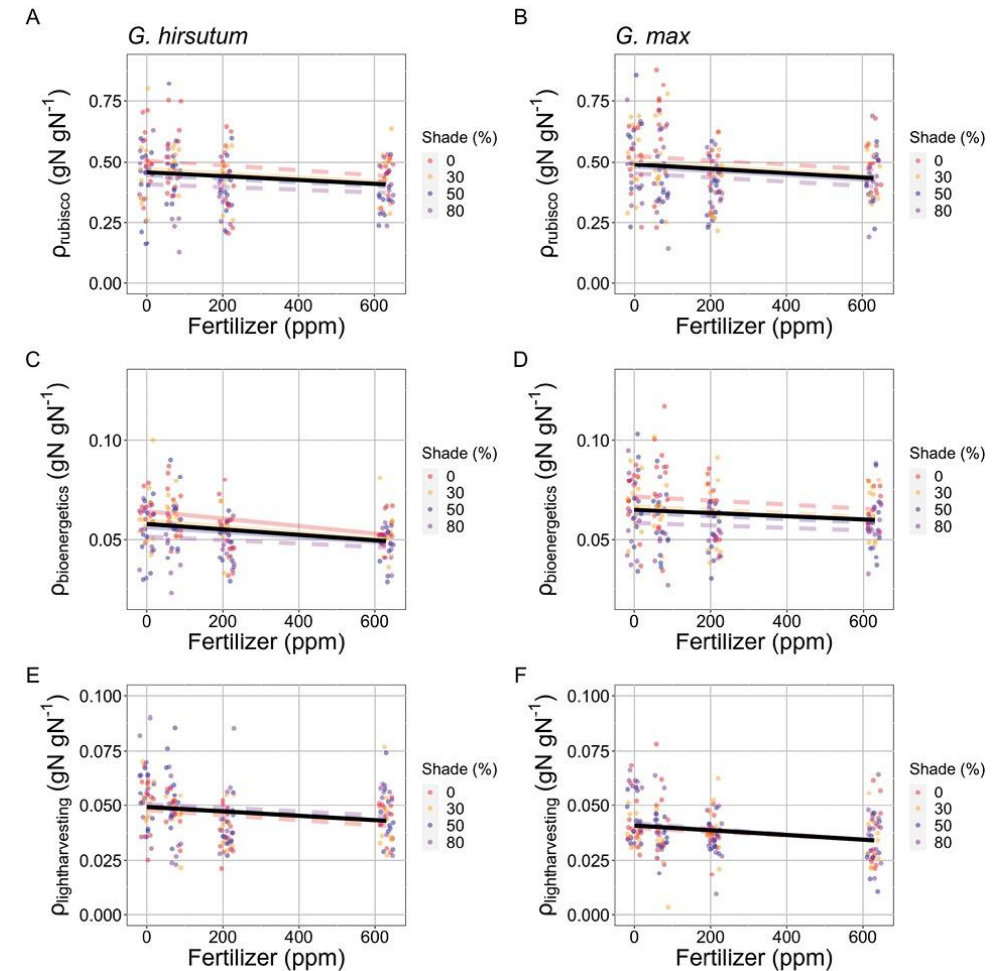
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Soil nitrogen impacts on allocation

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Soil N decreases leaf N allocation to photosynthesis



Soil phosphorus impacts on allocation

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